

8. This question is about momentum.

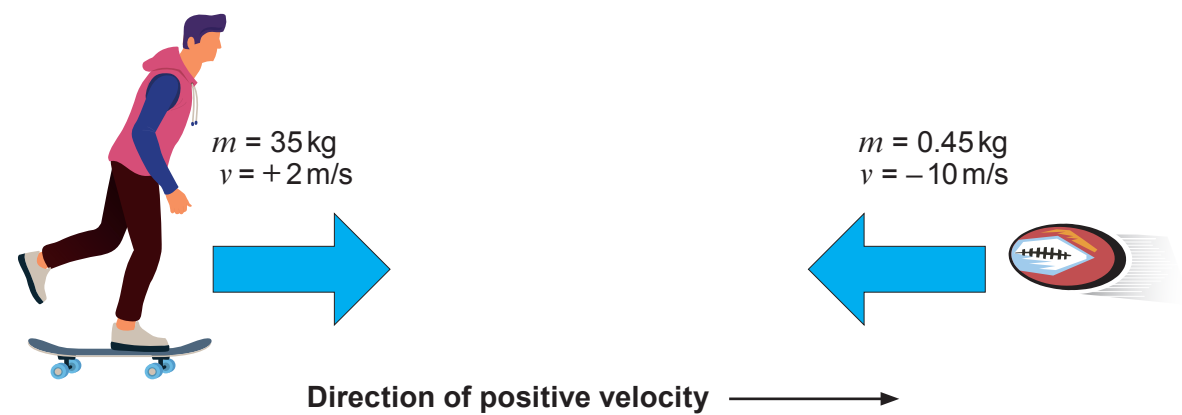
(a) State the principle of conservation of momentum. [2]

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(b) A skateboarder of mass 35 kg moving with a velocity of 2 m/s catches and holds on to a 0.45 kg rugby ball thrown directly towards him at 10 m/s.



Use the equation:

momentum = mass × velocity

to calculate the **total** momentum of the skateboarder and the ball before the collision and give the unit. [4]

Total momentum = .....

Unit = .....

